

STPA EXERCISE: LEVEL CROSSING

Last update: 05/04/2025

Exercise #1 – Read about the level crossing system on Wikipedia to get acquainted with the system to be analyzed in this exercise.

https://en.wikipedia.org/wiki/Level_crossing

Exercise #2 – Read the system description and the considerations for this exercise

System description

In the field of railway traffic control systems, a level crossing (also known as a grade crossing) is an intersection where a railway line crosses a road at the same level. It is estimated that, on average, 400 people in the European Union and over 300 people in the United States are killed at level crossings each year.

Modern trains have a much greater mass relative to their braking capability and therefore require a significantly longer braking distance than road vehicles. Road vehicles are more capable of avoiding accidents at level crossings than trains. One method to reduce such accidents is to provide “**passive protection**,” using warning signs, and “**active protection**,” using flashing lights, warning tones, and barriers or gates.

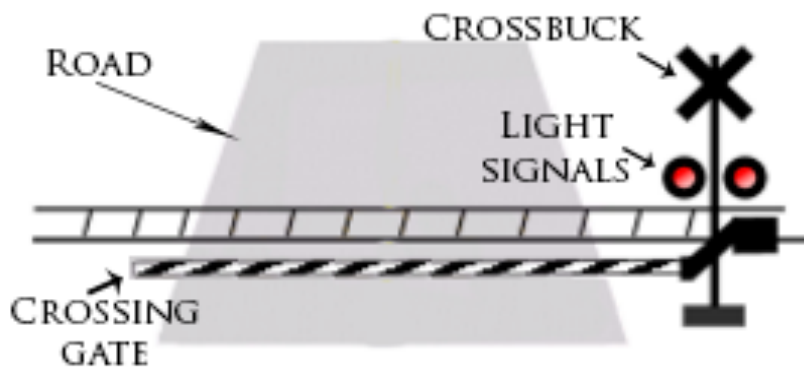


Figure 1. Illustrative image of a level crossing system

A crossbuck is a traffic sign used to indicate a railway level crossing. Crossbucks are sometimes supplemented by electrical warning systems such as flashing lights, a bell, or a boom barrier (also known as a boom gate or crossing gate) that descends to block the road and prevent traffic from crossing the tracks.

Considerations

For the system depicted in Figure 1, consider the following:

- The roadway is a one-way street, allowing traffic in only one direction.
- An automated controller closes the crossing gate when a train is approaching and opens it after the train has passed.

- A sensor is capable of detecting whether a train is approaching or has already passed.
- The crossing gate includes light signals but does not have sound signals.

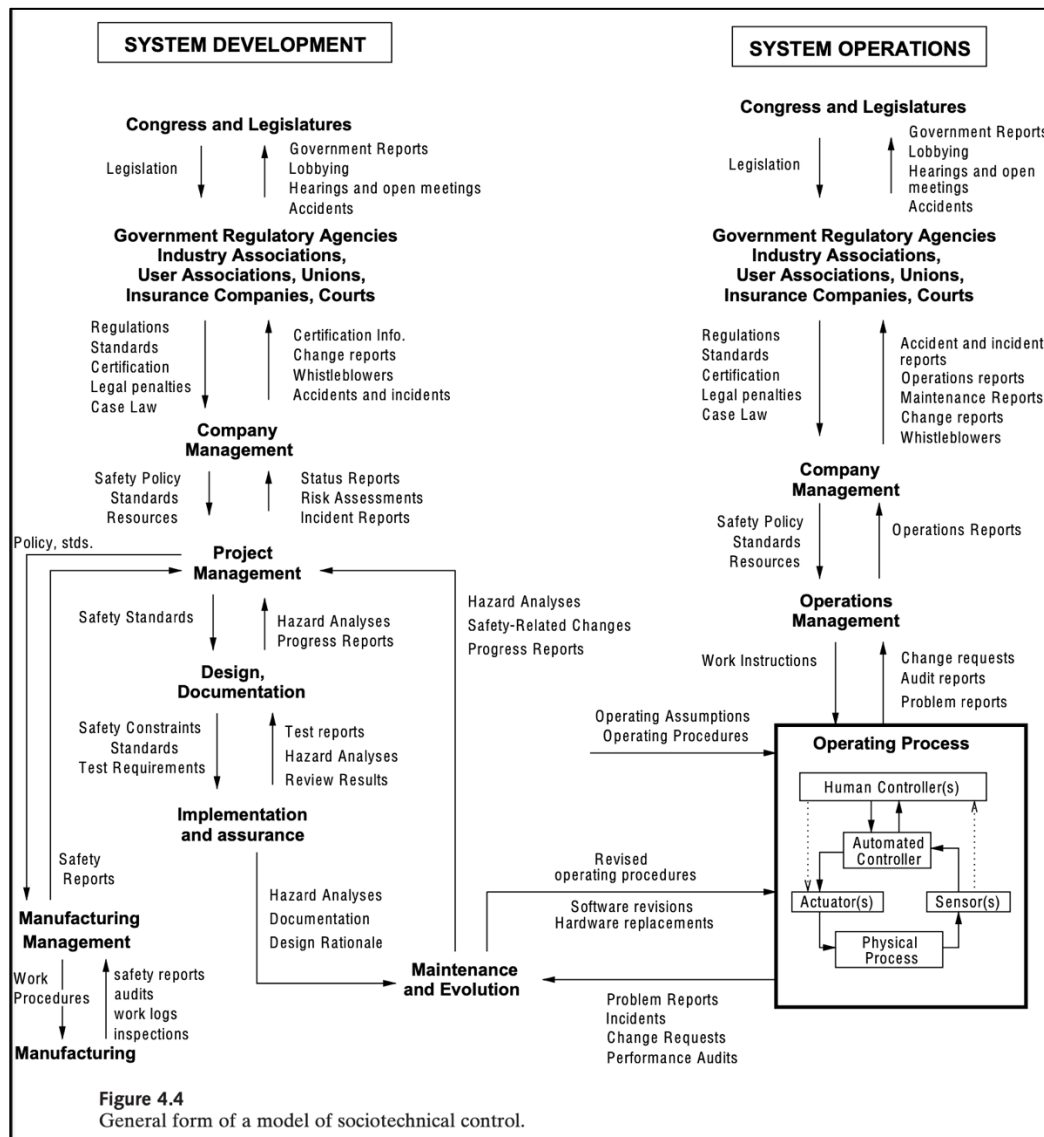
Exercise #3 – Perform the first step of STPA (define the purpose of the analysis) for the level crossing system.

Tip: You can find a template for the STPA analysis in the course site on Moodle (see the STPA section).

Exercise #4 – Perform the second step of STPA (model the control structure) for the level crossing system.

Tip #1: Before modeling the control structure, try to identify the components and their responsibilities needed to satisfy the safety constraints defined in Step 1. These responsibilities will help the analyst identify the control actions and feedback paths in the control structure.

Tip #2: Create two control structures: (i) a high-level control structure showing the development and operations hierarchies of components (see the image below, from *Engineering a Safer World*, for an example), and (ii) a detailed control structure focusing on the system under analysis.



Exercise #5 – Perform the third step of STPA (identify unsafe control actions) for the level crossing system.

Exercise #6 – Perform the fourth step of STPA (identify loss scenarios) for the level crossing system.

Exercise #7 – Read the abstract of the paper titled “An Automated Railway Level Crossing System.”

<https://bit.ly/3QJSg4a>

Exercise #8 – Extract all identified requirements (both functional and non-functional) from the STPA analysis. Use a separate document to record the requirements.

Tip: Watch again the video “*A Área de Conhecimento Requisitos de Software do SWEBOK Versão 4 (Parte 2 de 3)*”, which was discussed in the first week of the course. Pay special attention to the Requirements Specification work breakdown item (*Item Especificação de Requisitos na estrutura analítica de decomposição do SWEBOK*).

Good exercises !